SCIE 1120 Chemistry and Life

DNA replication and repair

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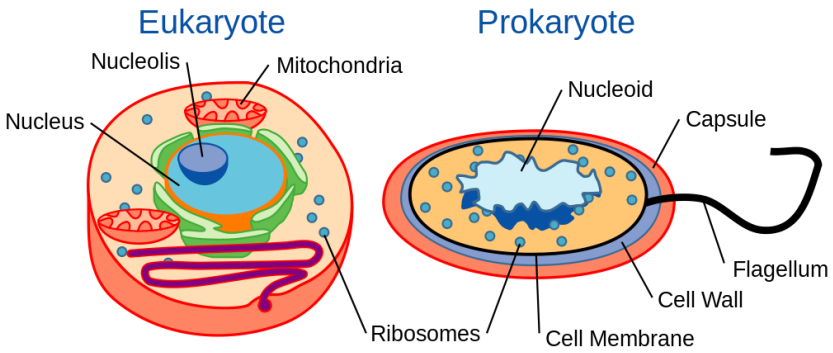
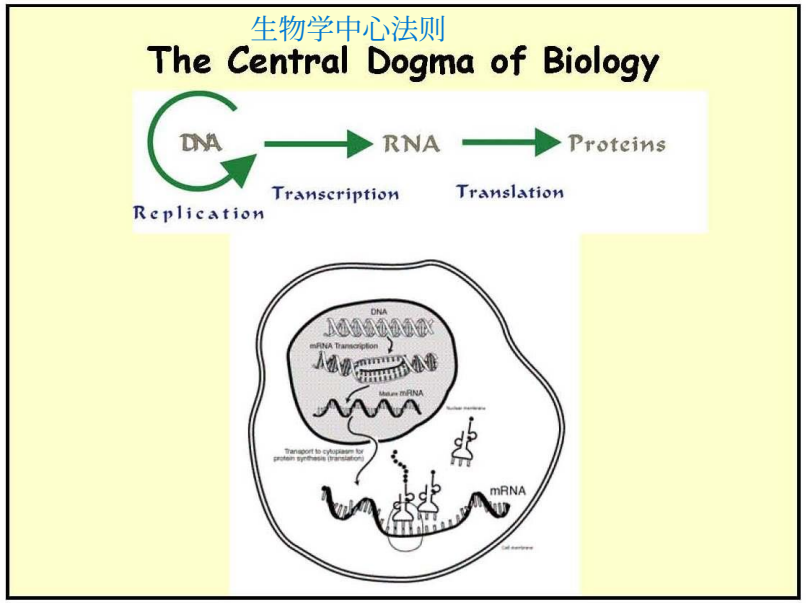
Apr.30, 2025, 3:00-4:20pm
LT-J



- All the DNA in a cell constitutes the cell's genome
- A genome can consist of a single DNA molecule (common in prokaryotic cells) or a number of DNA molecules (common in eukaryotic cells)



The eukaryotic cell's genetic instructions are housed in the nucleus and carried out by the ribosomes.

- The nucleus contains most of the DNA in a eukaryotic cell.
- Ribosomes use the information from the DNA to make proteins.

真核细胞的细胞核包含了大部分DNA
核糖体利用给来自DNA信息制造蛋白质

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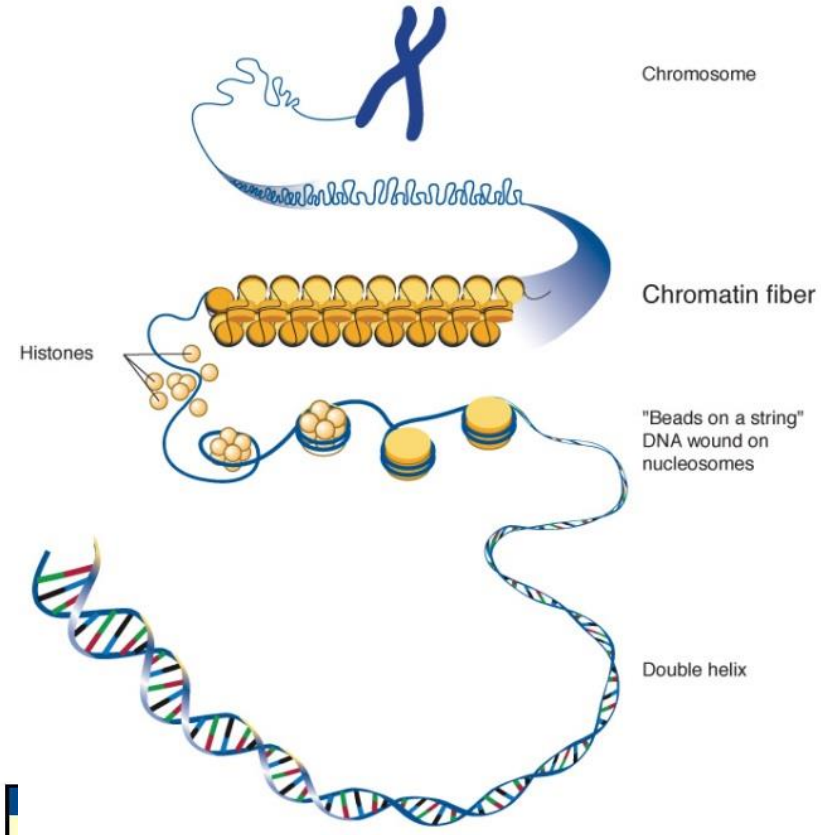
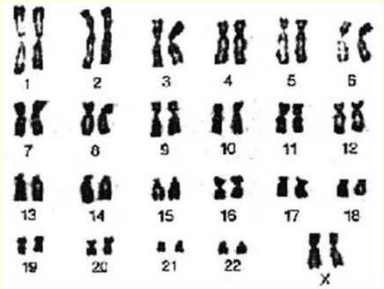


- In the nucleus, DNA is organized into discrete units called **chromosomes.** 染色体
- Each chromosome is composed of a single DNA molecule associated with proteins.
- The DNA and proteins of chromosomes are together called **chromatin.** 染色质
- Chromatin condenses to form discrete. **chromosomes** as a cell prepares to divide.
- The **nucleolus** is located within the nucleus and is the site of ribosomal RNA (rRNA) synthesis.

Human chromosomes and abnormality



Normal karyotype

Down syndrome karyotype

physical and mental developmental delays and disabilities

Ribosomes: Protein Factories

- **Ribosomes** are particles made of ribosomal RNA and protein.
- Ribosomes carry out protein synthesis in two locations:
  - In the cytosol (free ribosomes) 游离核糖体 cytosol细胞质
  - On the outside of the endoplasmic reticulum or the nuclear envelope (bound ribosomes) 内质网 核膜 结合核糖体

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The extra genetic material present in Down syndrome results in overexpression of a portion of the 310 genes located on chromosome 21

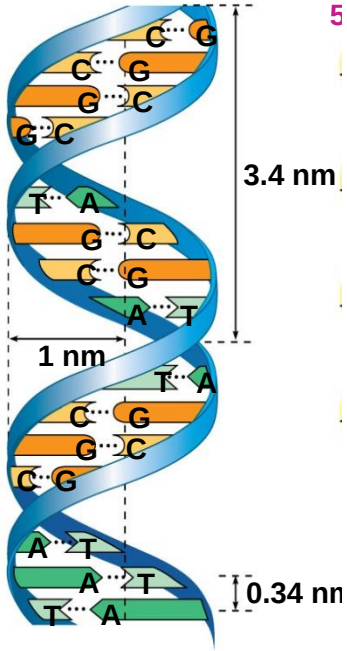
The rate (norm 1/1000) increase in stress levels seen in couples during the time of conception

Also, XXX syndrome, trisomy X

DNA facts

Human Genome

三十亿碱基对



(a)Key features of DNA structure

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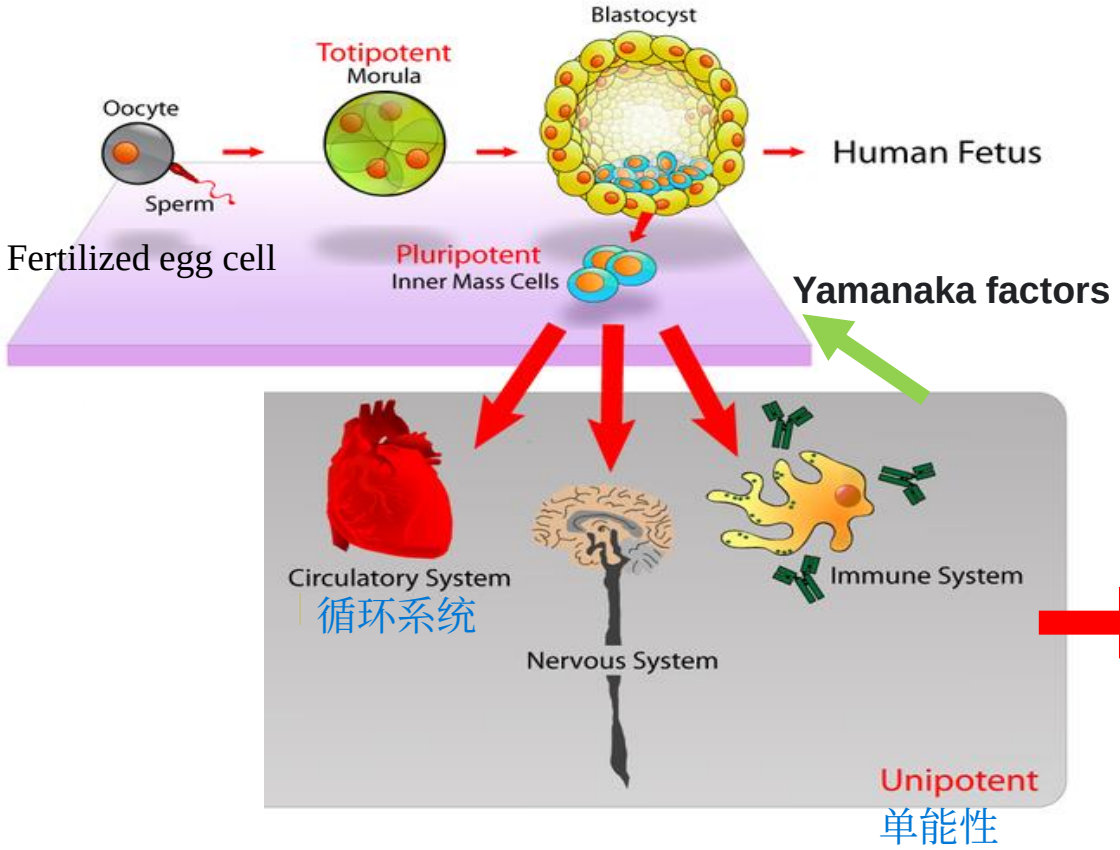
- The human genome comprises (23K genes) the information contained in one set of human chromosomes which themselves contain about **3 billion base pairs** (bp) (1.5% for genes) of DNA in **46 chromosomes** (22 autosome pairs (not sex chromosomes)+ 2 sex chromosomes). The total length of DNA present in one adult human is calculated by the multiplication of

  (length of 1 bp)(number of bp per cell)(number of cells in the body)

  $(0.34 \times 10^{-9}\ \text{m})(6 \times 10^{9})(10^{14}) = 2.0 \times 10^{14}$ meters

- That is the equivalent of nearly 600 trips from the earth to the sun and back.
- There are 100 trillion (10^{14}) cells in your body.
- If unwound and linked together, the strands of DNA in one cell would stretch almost **six feet (2Meters)**

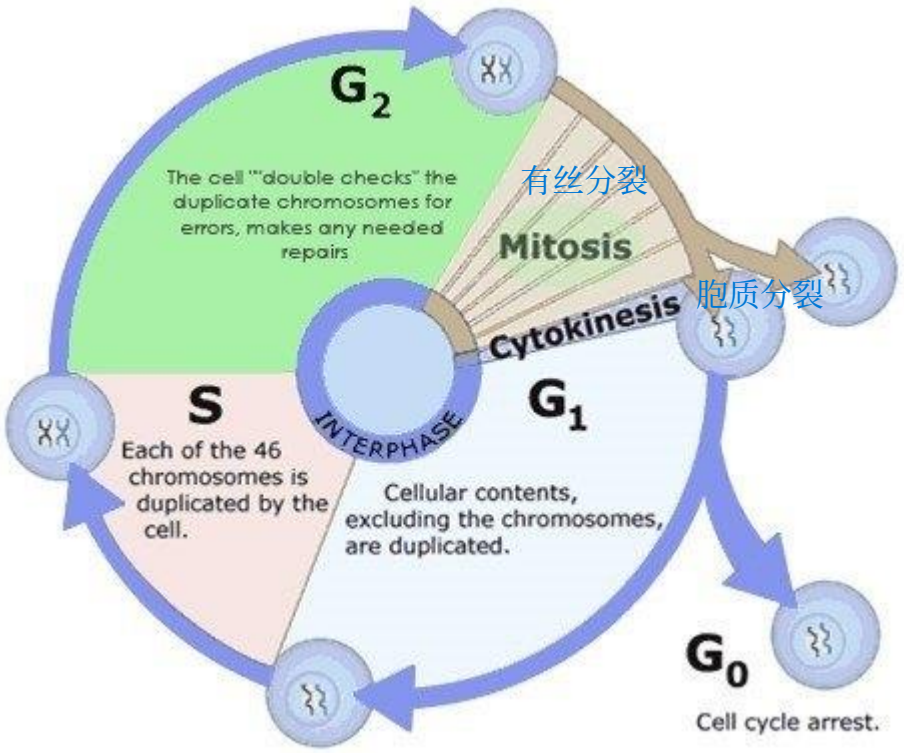
Human development



100 billion (10^9) neurons
20% energy consumption

100 trillion (10^{12}) cells

Cell cycle

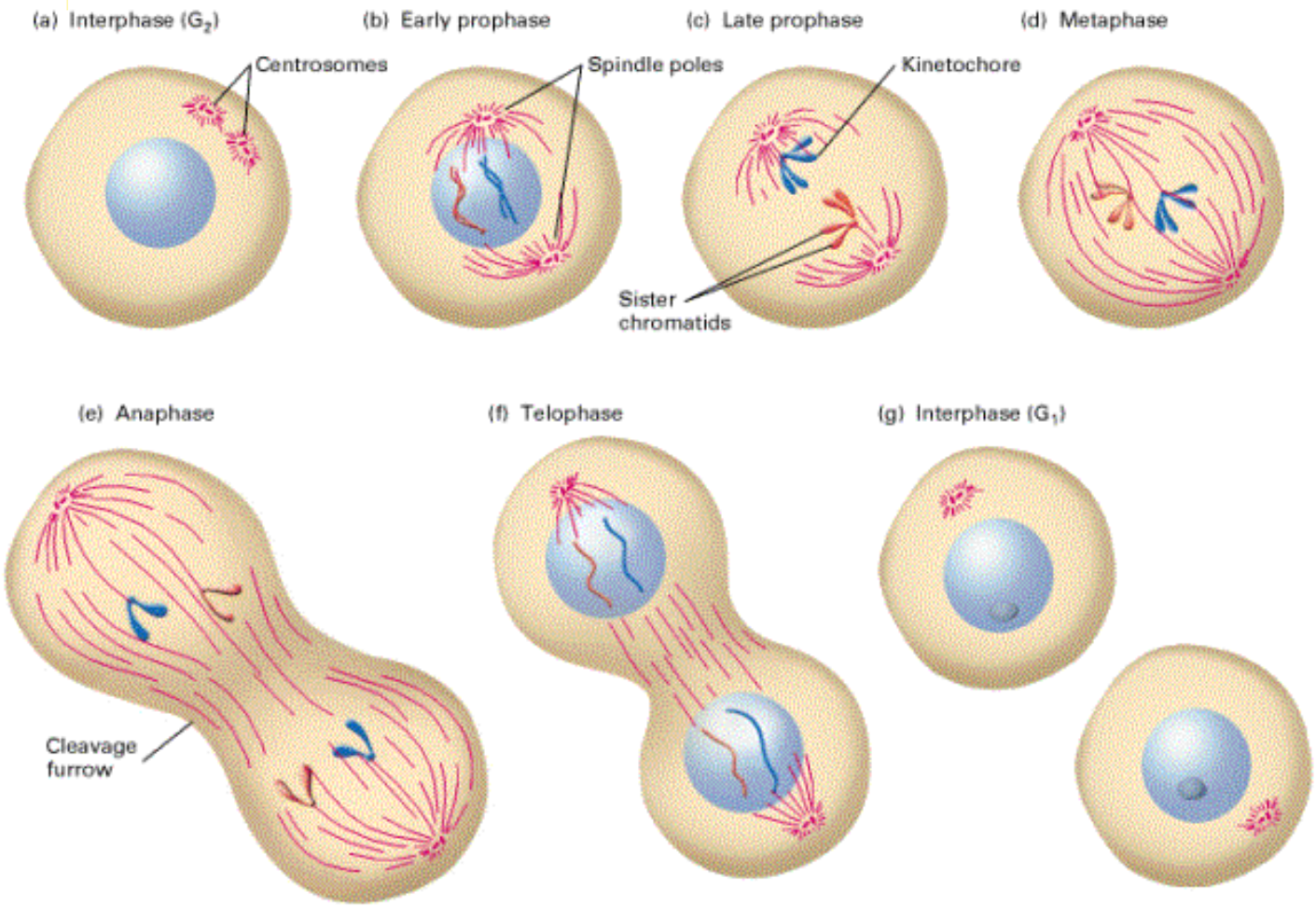


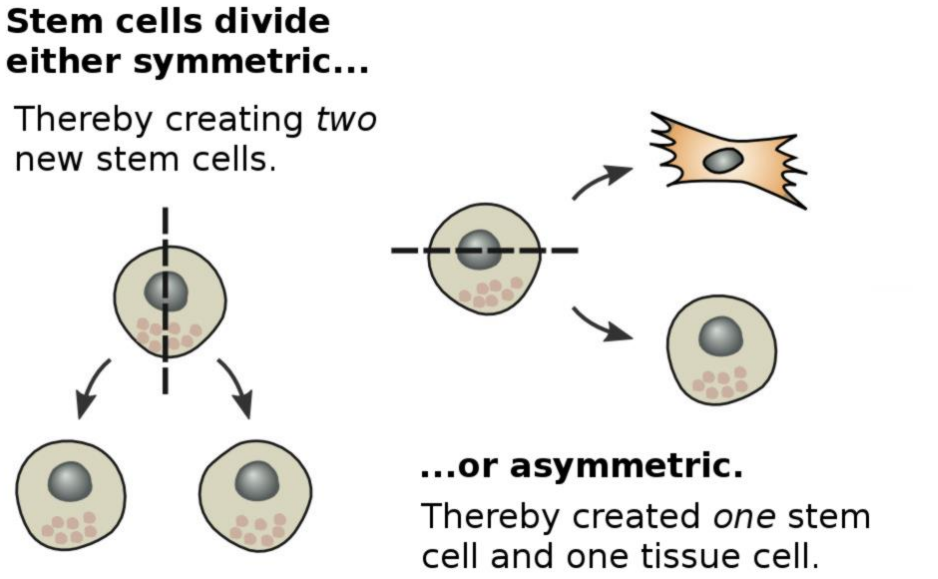
Human cell cycle

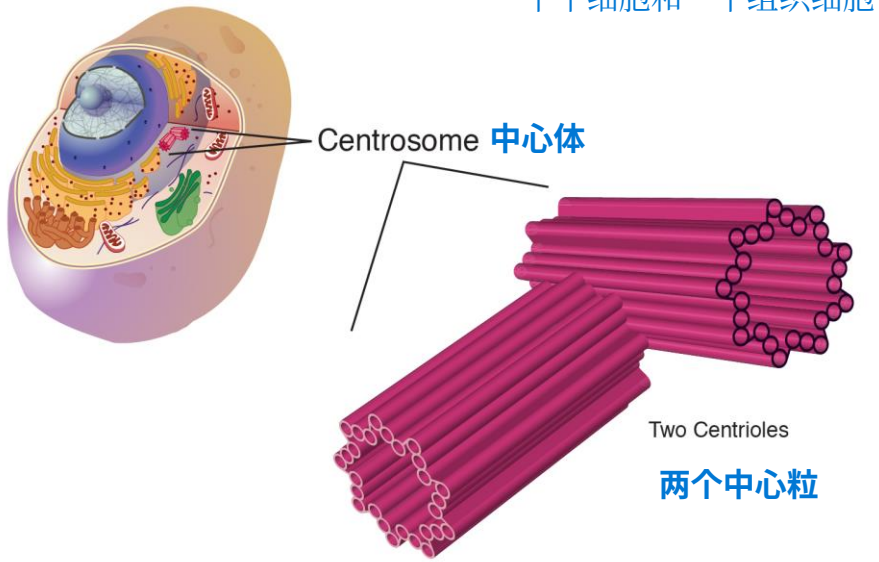
Cell cycles, Chromosomes & Mitosis.mp4

- **G1**: Cells increase in size in Gap 1, produce RNA and synthesize protein ensures that everything is ready for DNA synthesis.
- **S Phase**: To produce two similar daughter cells, the complete DNA instructions in the cell must be duplicated. DNA replication occurs during this S (synthesis) phase.
- **G2**: During this gap between DNA synthesis and 有丝分裂 mitosis, the cell will continue to grow, produce new proteins and make needed repairs. Preparing for cell division (Cytokinesis)胞质分裂
- **M Phase**: Cell growth and protein production stop at this stage in the cell cycle. All of the cell's energy is focused on the complex and orderly division into two similar daughter cells.
- **G0**: Quiescent 静止非活跃期 (inactive) stage. Some cells in the adult animals do not appear to exhibit division (e.g., heart cells) and many other cells divide only occasionally, as needed to replace cells that have been lost because of injury or cell death.
- **Cytokinesis:** 细胞质分裂 It usually ends with division of cytoplasm. 细胞质
- The duration of these cell cycle phases varies considerably in different kinds of cells. For a typical rapidly proliferating 快速增殖 human cell with a total cycle time of 24 hours, the G_1 phase might last about 11 hours, S phase about 8 hours, G_2 about 4 hours, and M about 1 hour. ~95% cells are replaced in 7 years.

mitosis

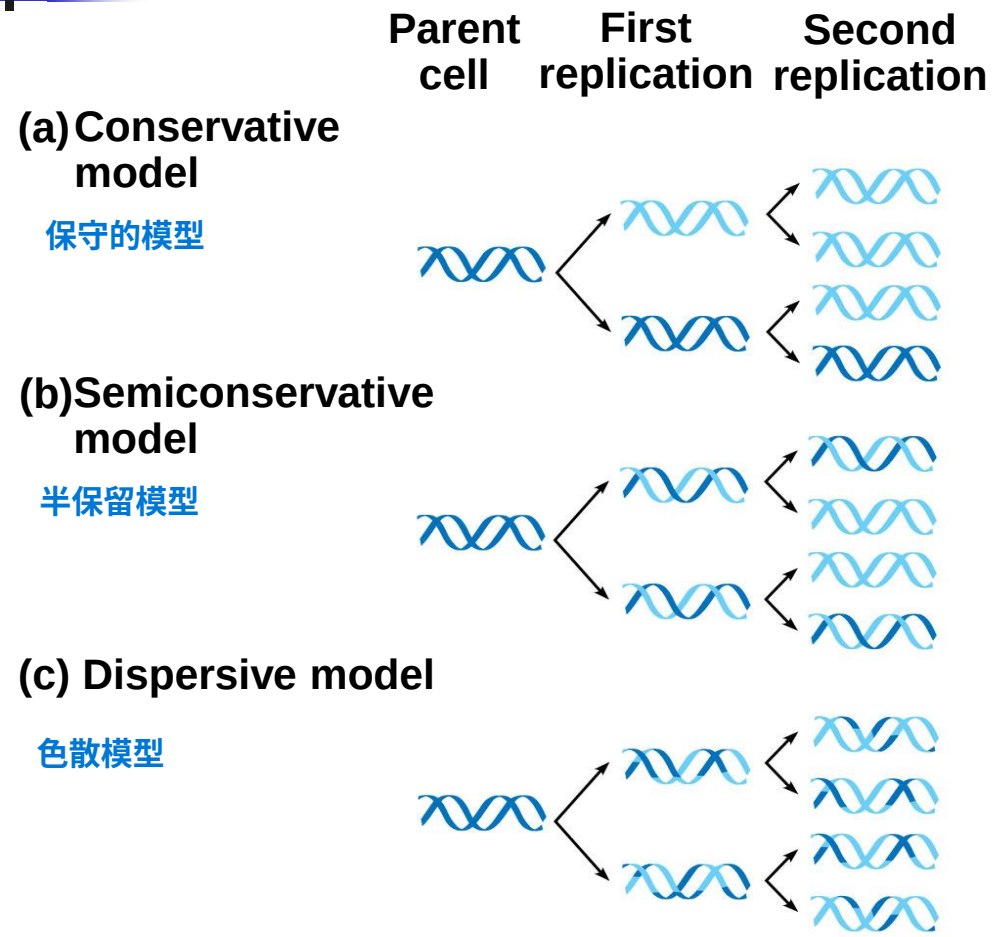






Actual Footage of Cell Division (Kidney Cells).mp4

DNA replication



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- In DNA replication, the parent molecule 解旋 unwinds, and two new daughter strands are built based on base-pairing rules (AT, GC)
- Watson and Crick's **semiconservative** 半保留复制模型 **model** of replication predicts that when a double helix replicates, each daughter molecule will have one old strand (derived or "conserved" from the parent molecule) and one newly made strand
- Competing models were the conservative model (the two parent strands rejoin) and the dispersive model (each strand is a mix of old and new) **保守模型**（两条亲本链重新连接）**分散模型**（每条链都是新旧片段的混合）

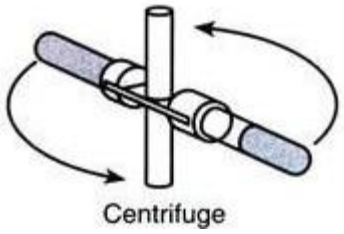
Meselson and Stahl Experiment

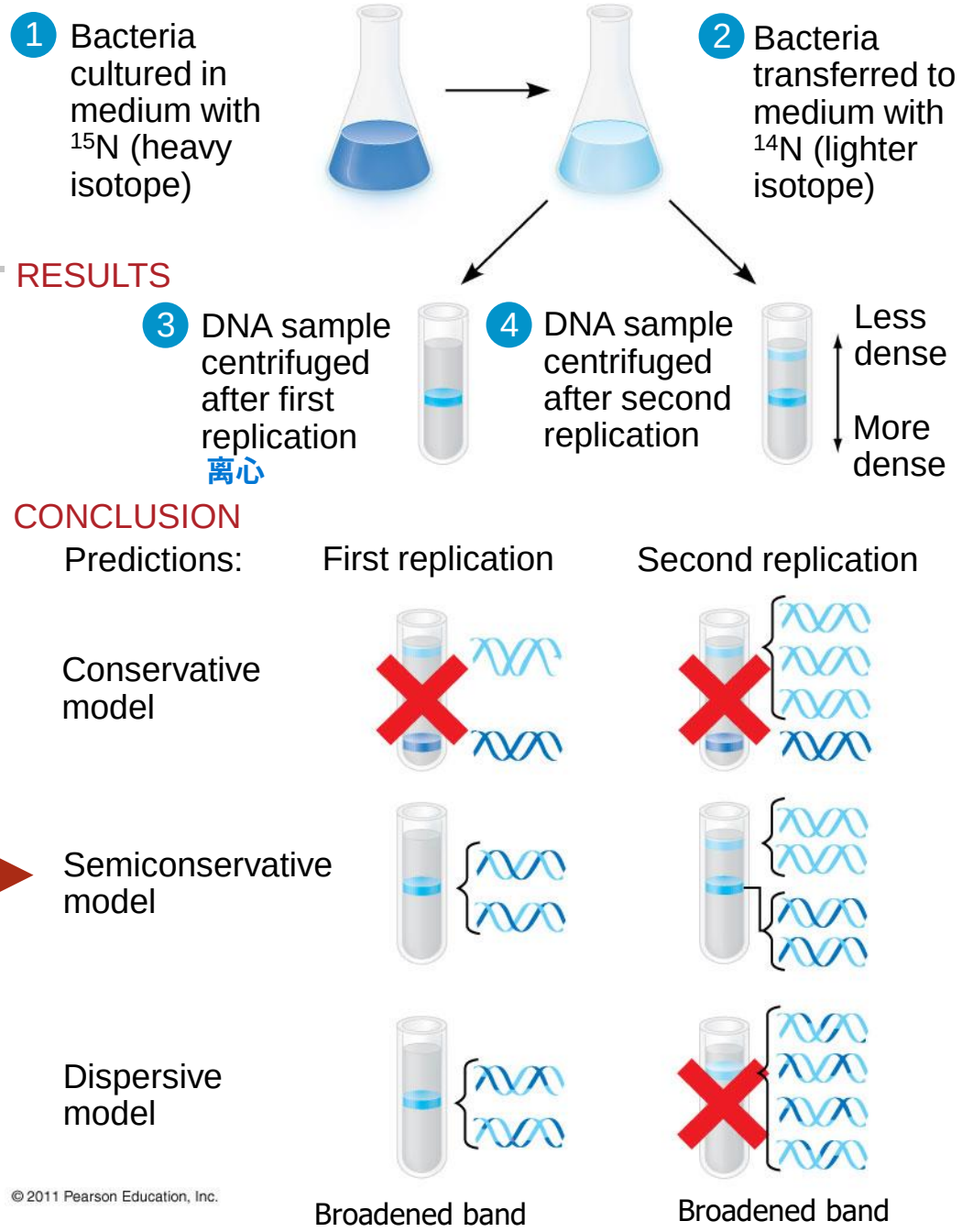
Does DNA replication follow the conservative, semiconservative, or dispersive model?

Experiments by Matthew Meselson and Franklin Stahl supported the semiconservative model

(1) All DNAs in Bacteria (4.7M bps) are labeled with 15N (original strands)

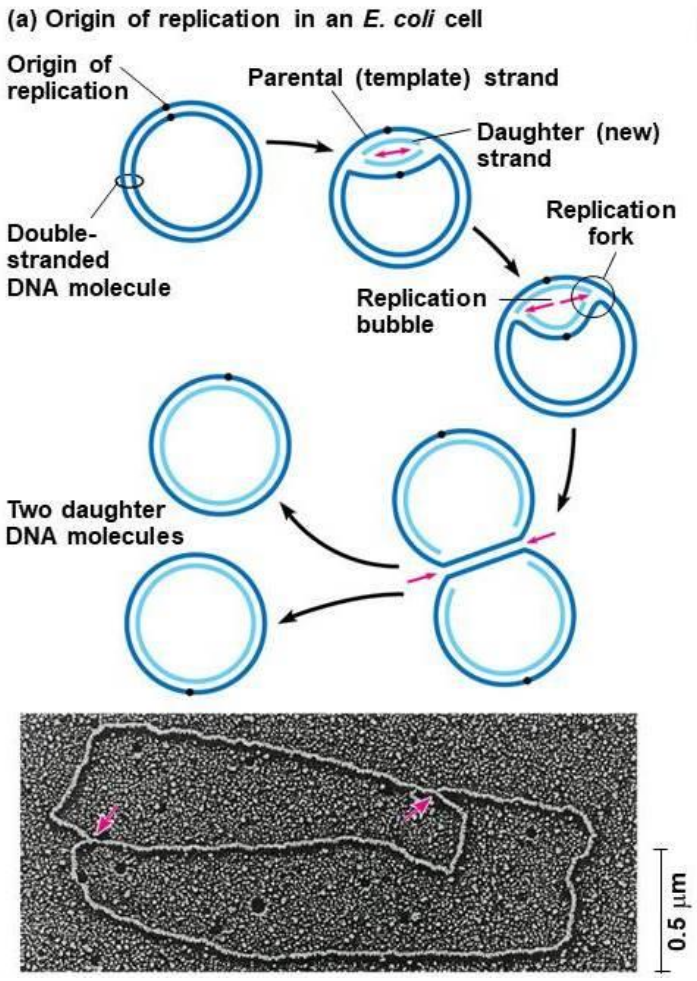
(2) New strands are labeled with 14N (E. Coli: about 40mins, 4.7M bp (1000bp/s))

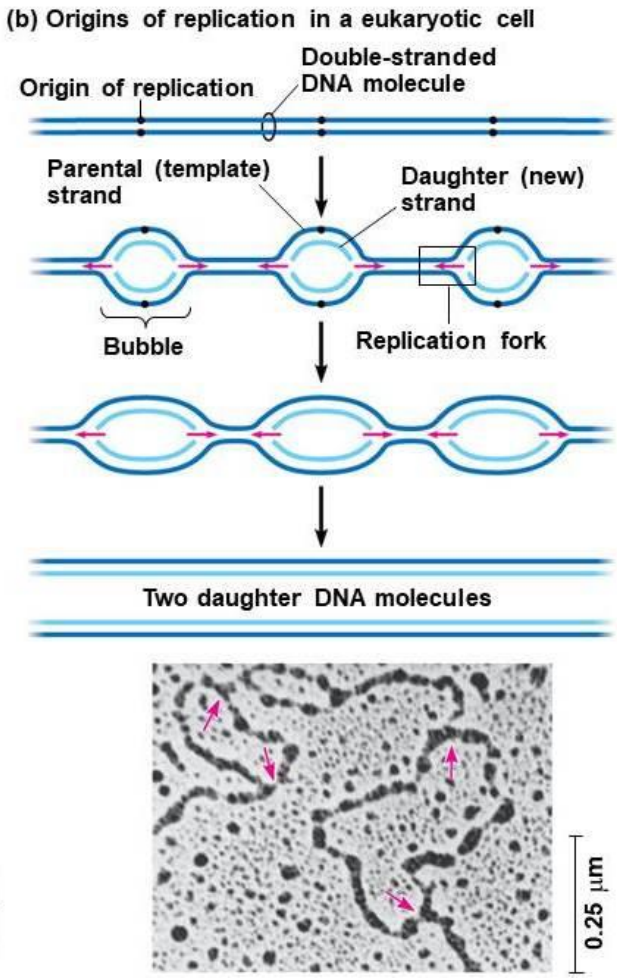




DNA Replication: *A Closer Look*

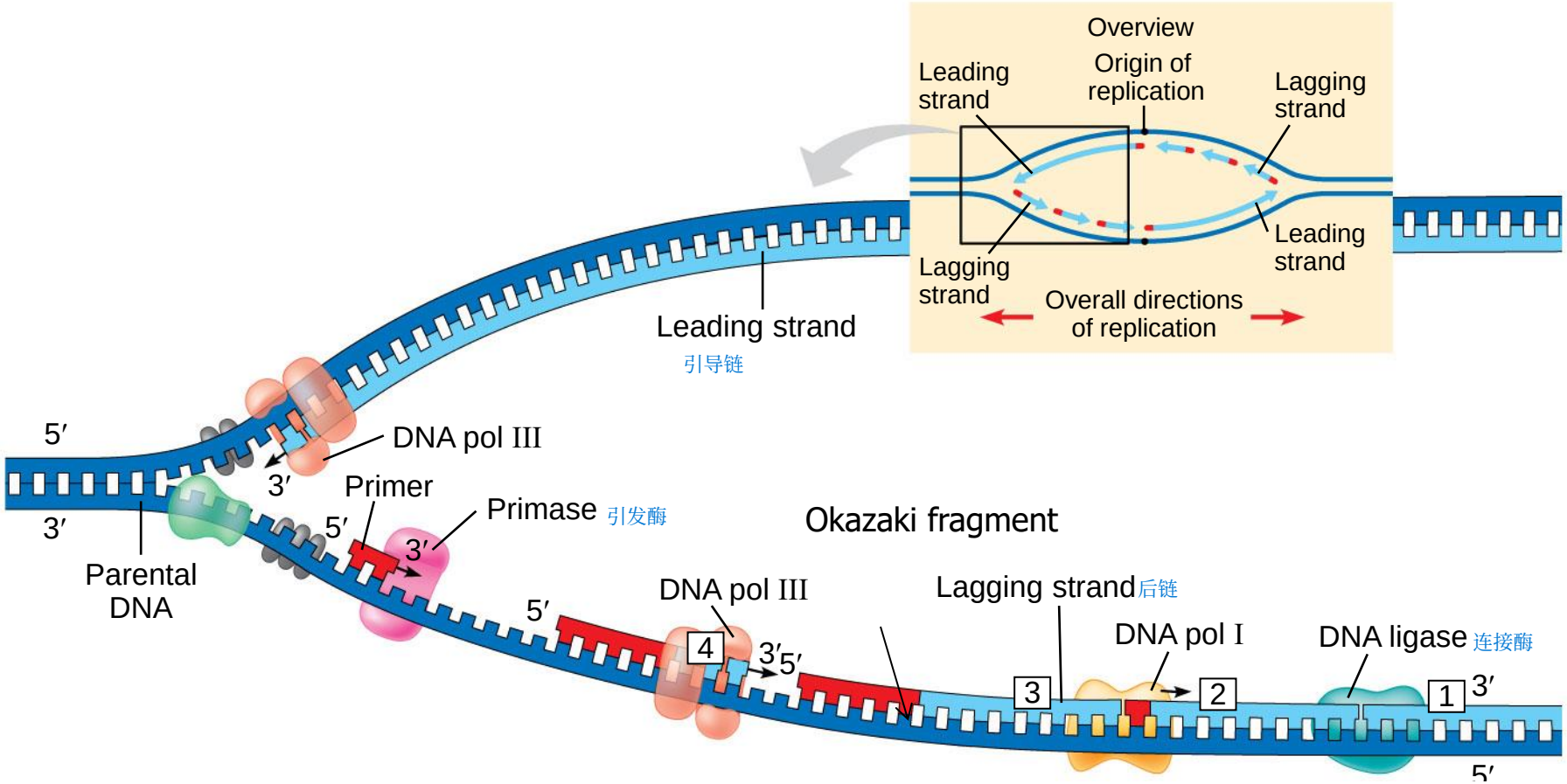
真核细胞复制的起源



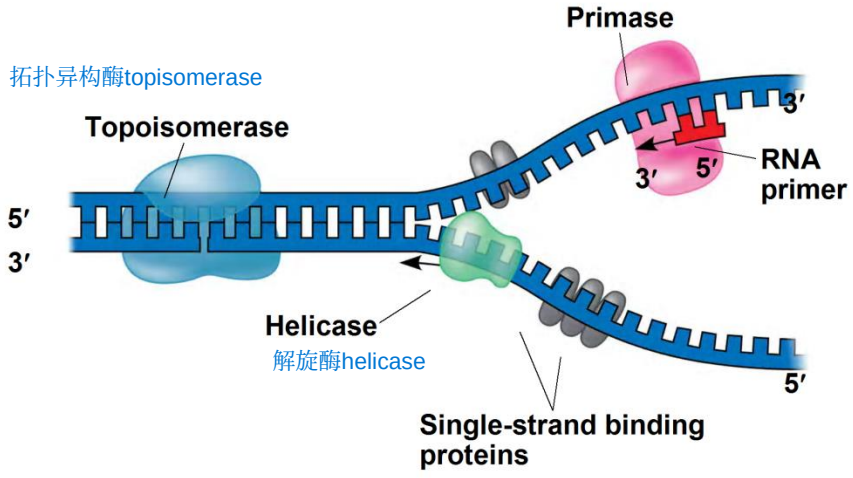


- The copying of DNA is remarkable in its speed and accuracy (one error in 10^7 bps)
- More than a dozen enzymes and other proteins participate in DNA replication
- Replication begins at particular sites called **origins of replication**, where the two DNA strands are separated, opening up a replication "bubble" by a DNA helicases 解旋酶 binding on a 200bps origin sequence
- A eukaryotic chromosome may have hundreds or even thousands of origins of replication
- Replication proceeds in both directions from each origin, until the entire molecule is copied

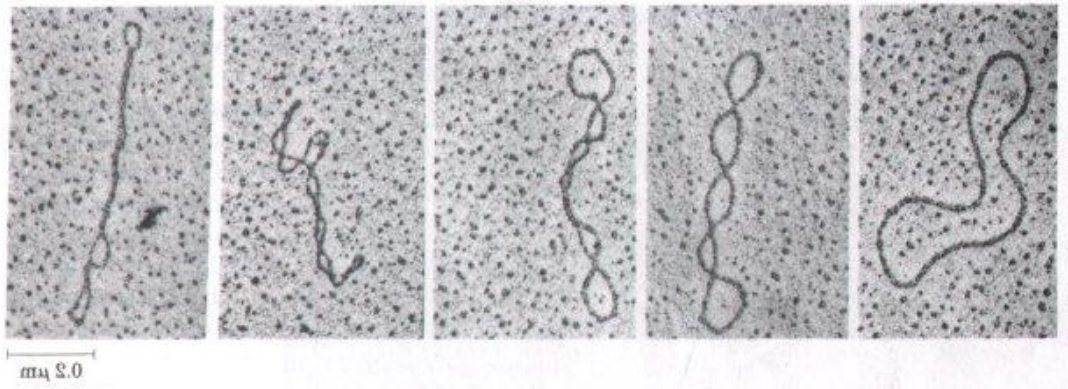
Prokaryotic DNA Replication: *A Closer Look*



- DNA replication is facilitated by proteins.
- The copying of DNA is remarkable in its speed and accuracy
- More than a dozen enzymes and other proteins participate in DNA replication
- Initiation, elongation and termination **起始，延伸，**终止
- GTC with G/T-rich flanks (侧翼序列), which increase primase-DNA binding up to 10-fold for primase (iScience 2018)

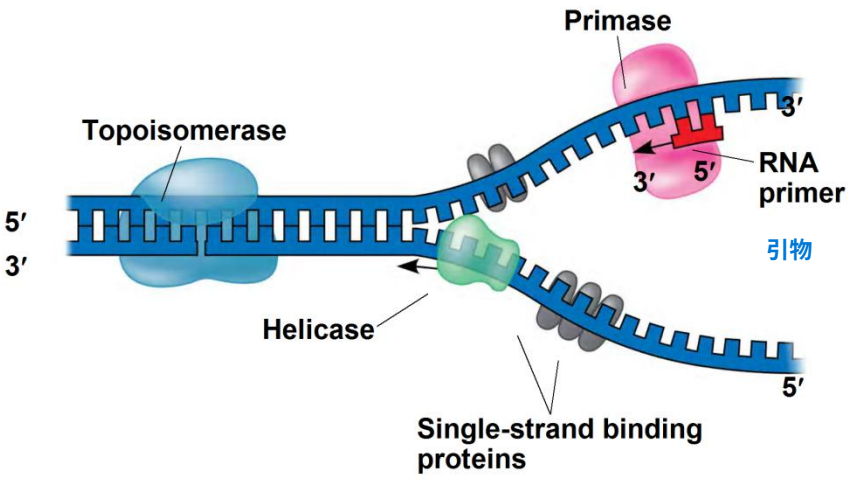


relaxed- supercoiled DNA



- At the end of each replication bubble there is a **replication fork**, a Y-shaped region where new DNA strands are elongating 延伸
- 解旋酶 **Helicases** are enzymes that untwist the double helix at the replication forks (origins) and break H-Bonds
- **Single-strand binding proteins** bind to and stabilize single-stranded DNA preventing structure formation and nuclease cutting.
- **Topoisomerase** corrects "overwinding" 过度缠绕 ahead of replication forks by breaking, swiveling, and rejoining DNA strands to release 旋转 the tension of the DNA
- 聚合酶 DNA polymerases cannot initiate synthesis of a polynucleotide; they can only add nucleotides to the 3′ end 多核苷酸
- The initial nucleotide strand is a short RNA **primer** as DNA polymerase can only add nucleotide from 3′ end.

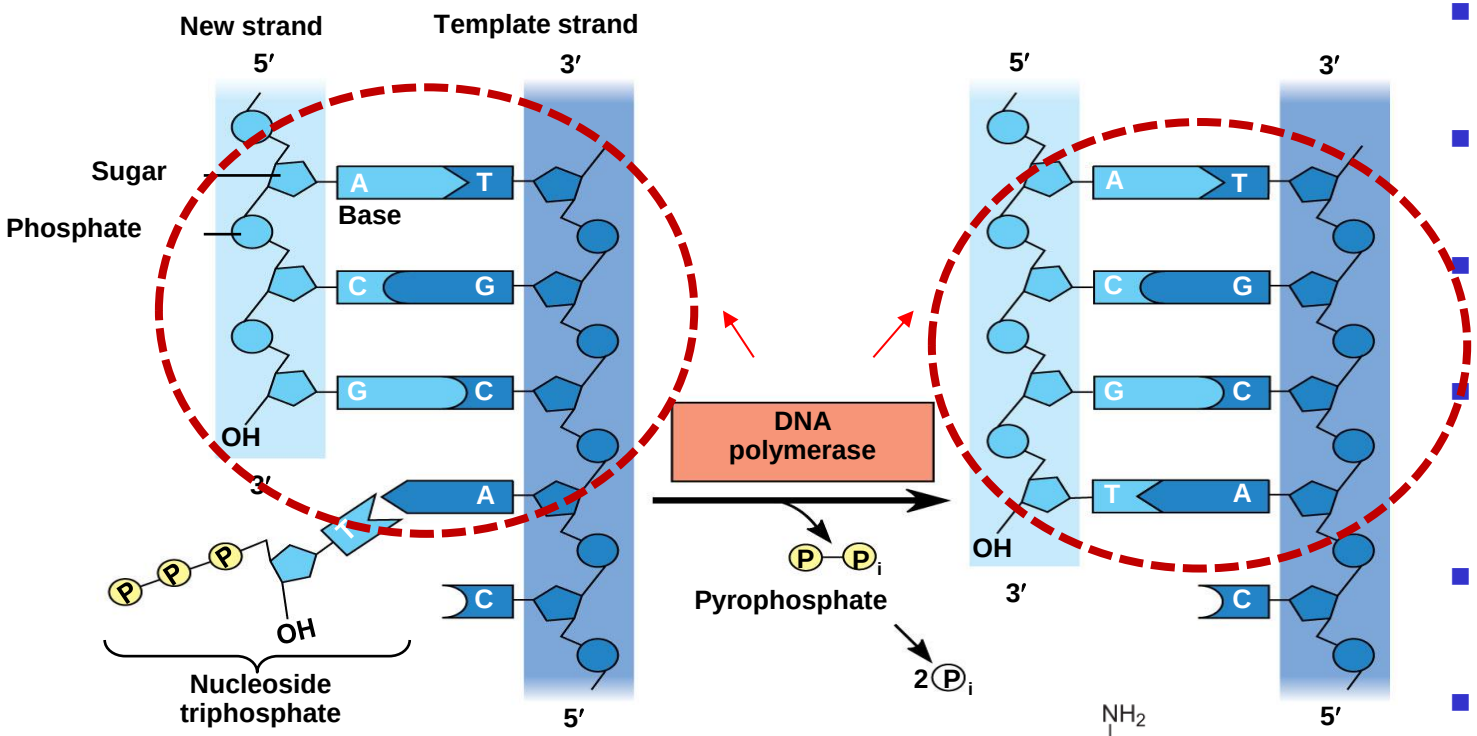
Prokaryotic DNA Replication: *A Closer Look*



有一种叫做引物酶的酶可以从零开始合成 RNA 链，并以亲本 DNA 为模板，一次添加一个 RNA 核苷酸。
引物很短（5 到 10 个核苷酸长），其 3' 端是新 DNA 链的起始点。
被称为 DNA 聚合酶的酶在复制叉处催化新 DNA 的合成，其出错率低于 10 的负 7 次方。
大多数 DNA 聚合酶需要引物和 DNA 模板链。
细菌的延伸速率约为每秒 1000 个核苷酸，而人类细胞中约为每秒 50 个核苷酸。
在原核细胞中，当两个复制泡相遇时 DNA 复制就会停止（因为原核细胞（细菌）中的基因组 DNA 是环状的）。而在真核细胞中，基因组 DNA 是线性的，就会出现“末端问题”

- An enzyme called **primase** can start an RNA chain from scratch and adds RNA nucleotides one at a time using the parental DNA as a template
- The primer is short (5–10 nucleotides long), and the 3′ end serves as the starting point for the new DNA strand
- Enzymes called **DNA polymerases** catalyze the elongation of new DNA at a replication fork with an error rate < 10^{7}
- Most DNA polymerases require a primer and a DNA template strand
- The rate of elongation is about 1000 nucleotides per second in bacteria and 50 per second in human cells
- DNA replication stops when two bubbles meet in prokaryotic cells (as genomic DNA in prokaryotic cell (Bacteria) are circular). In Eukaryotic cells, genomic DNA is linear, there will be a “end problem”.

Prokaryotic DNA Replication: *A Closer Look*



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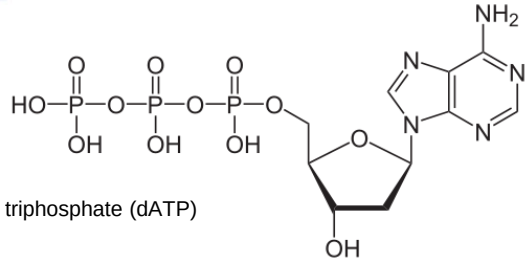
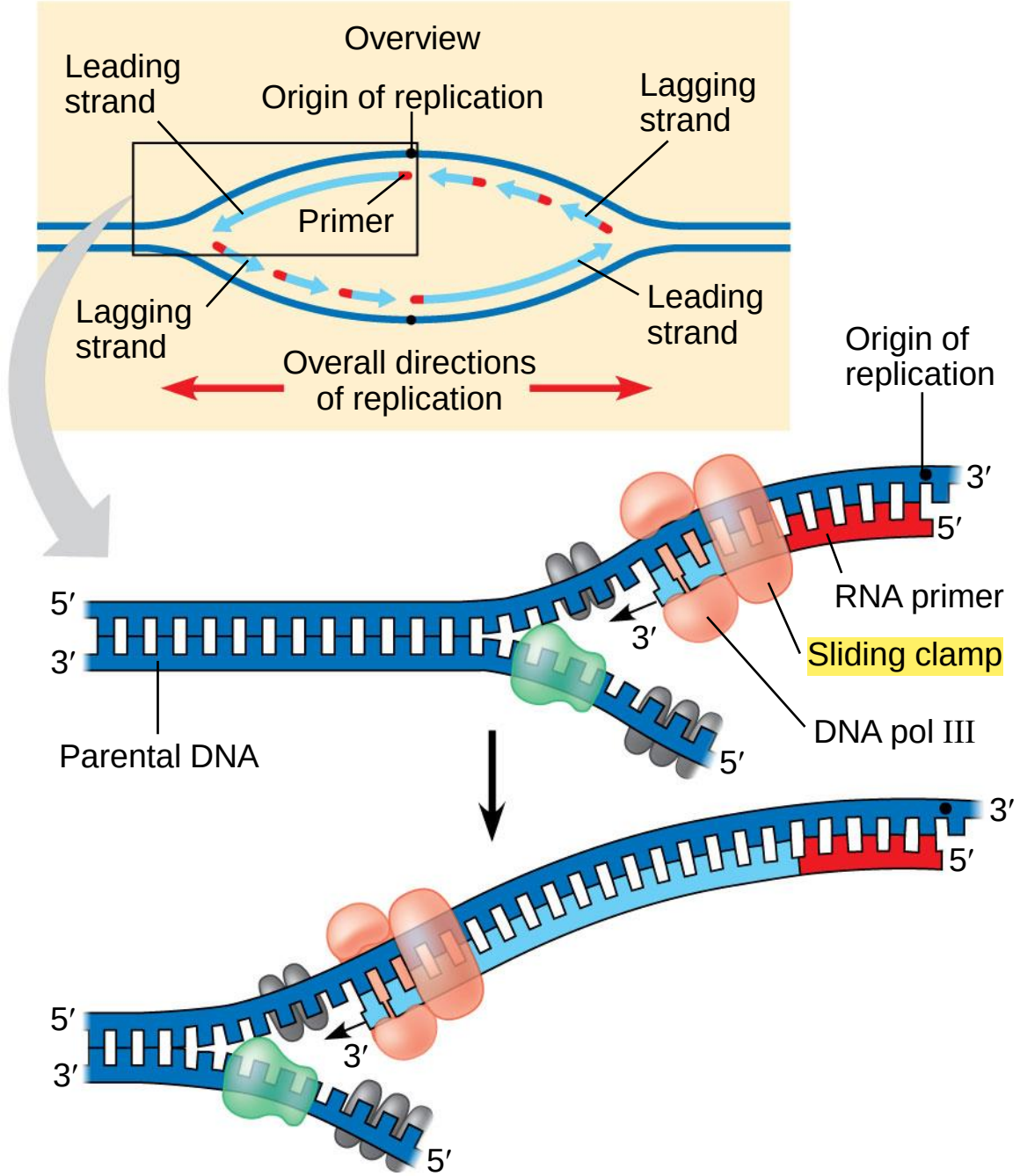
dNTP



Deoxyadenosine triphosphate (dATP)

- Each nucleotide that is added to a growing DNA strand is a **nucleoside triphosphate**
- dATP supplies adenine to DNA and is similar to the ATP of energy metabolism
- The difference is in their sugars: dATP has deoxyribose while ATP has ribose
- As each monomer of dNTP joins the DNA strand, it loses two phosphate groups as a molecule of pyrophosphate
- The antiparallel structure of the double helix affects replication
- DNA polymerases add nucleotides only to the free 3′ end of a growing strand; therefore, a new DNA strand can elongate only **in the 5′ to 3′ direction**
- DNA polymerase gets its energy from dNTPs

- Along one template strand of DNA, the DNA polymerase synthesizes a **leading strand** continuously, moving toward the replication fork
- To elongate the other new strand, called the **lagging strand**, DNA polymerase must work in the direction away from the replication fork
- The lagging strand is synthesized as a series of segments called **Okazaki fragments** (~1000bps), which are joined together by **DNA ligase** 连接酶
- The **sliding clamp** encircles the DNA and allows the polymerase to slide easily along the DNA template strand during replication.



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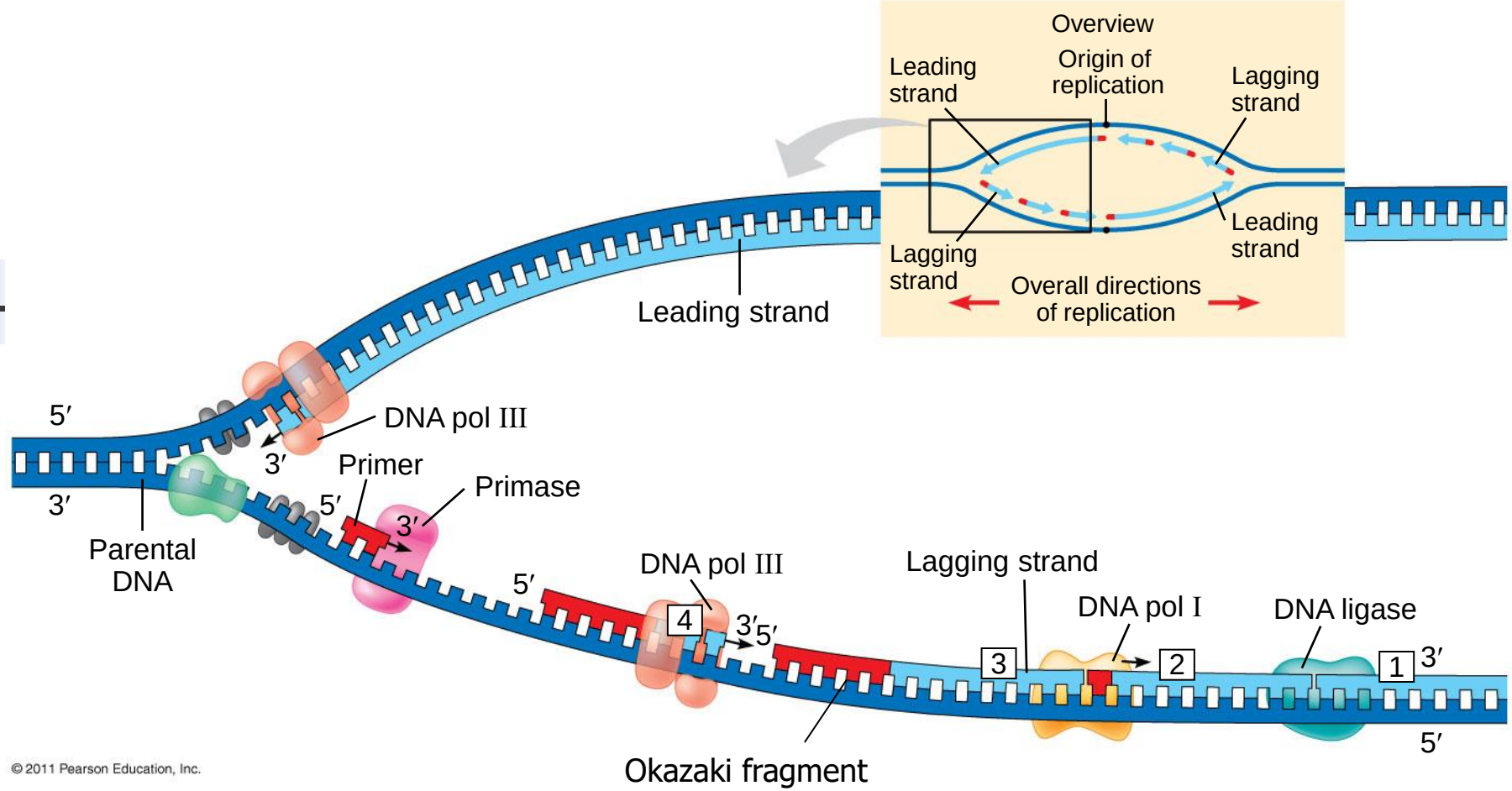


- **DNA Pol I** is mainly to repair any damage with DNA, but it also serves to connect Okazaki fragments by deleting RNA primers and replacing the strand with DNA in prokaryotic cells

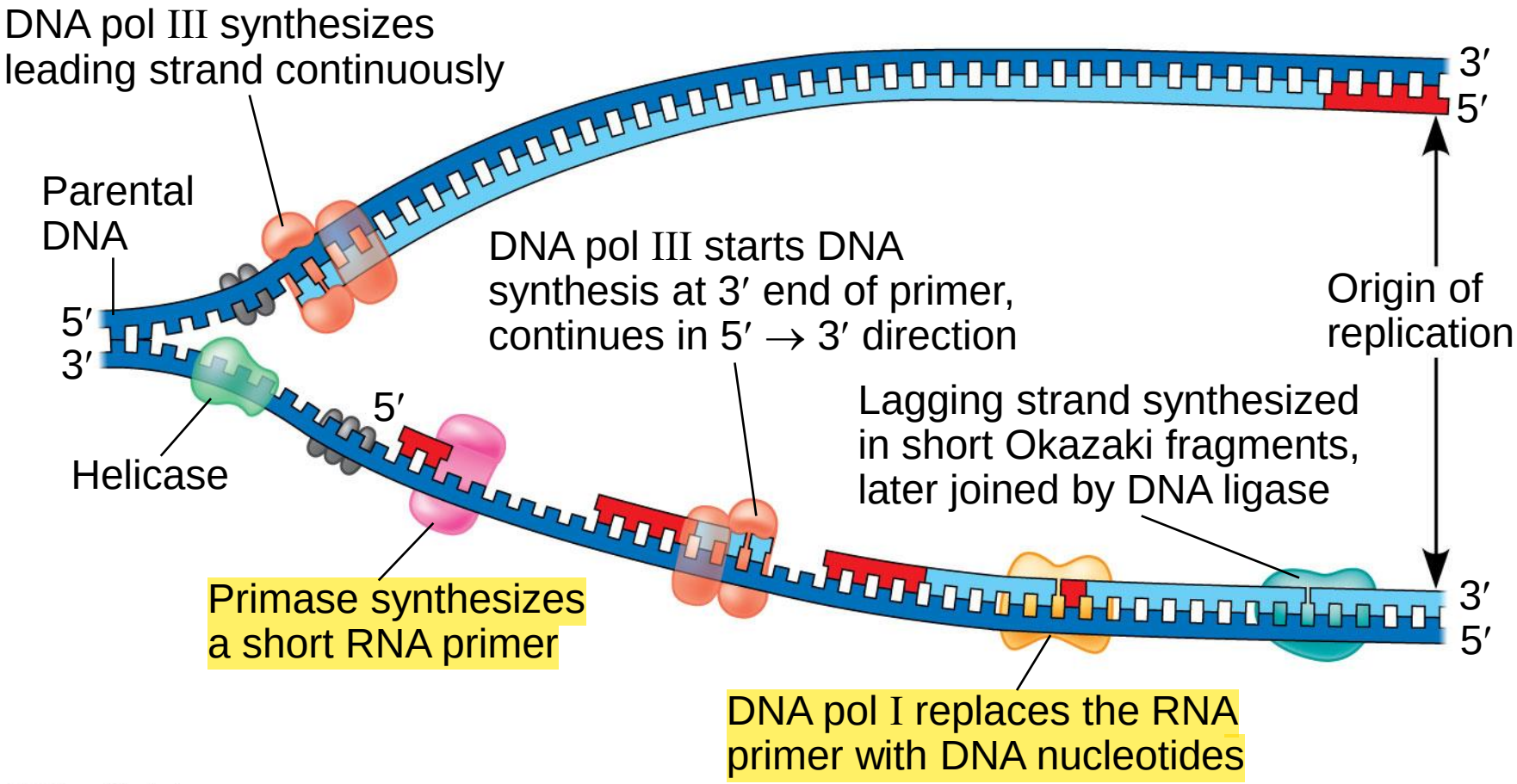
  DNA 聚合酶 I **主要用于修复** DNA 的任何损伤，但在原核细胞中，它还通过切除 RNA **引物并用** DNA 替换该链来连接冈崎片段。

- **DNA Pol III** is an enzyme used for both leading and lagging strand synthesis. is the <u>primary enzyme</u> complex involved in prokaryotic DNA replication. the DNA Pol III holoenzyme also has proofreading capabilities that corrects replication mistakes by means of exonuclease activity reading 3'→5' and synthesizing 5'→3'. in prokaryotic cells (error <10^{9})

  外切核酸酶

  DNA 聚合酶 III 用于前导链和滞后链的合成。是原核生物 DNA 复制的主要酶复合物。DNA 聚合酶 III 全酶还具有校对功能，通过 3-5的外切核酸酶活性和5-3的合成活性来纠正复制错误。

Review of DNA Replication



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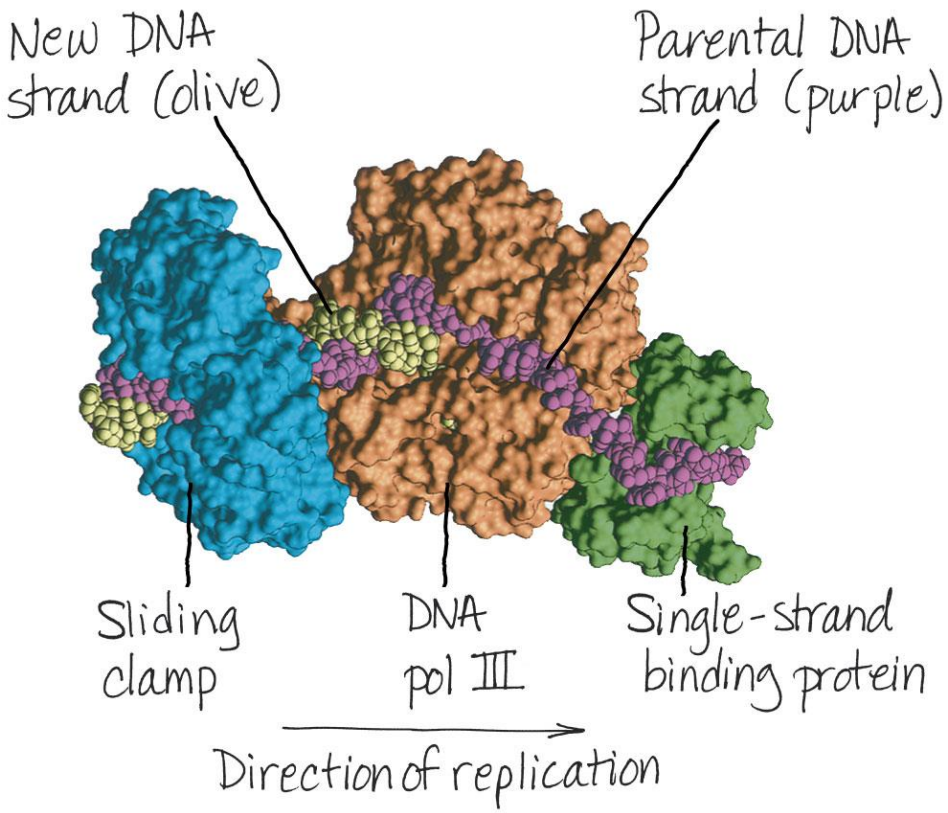
DNA Replication [HD Animation].mp4

The DNA Replication Complex



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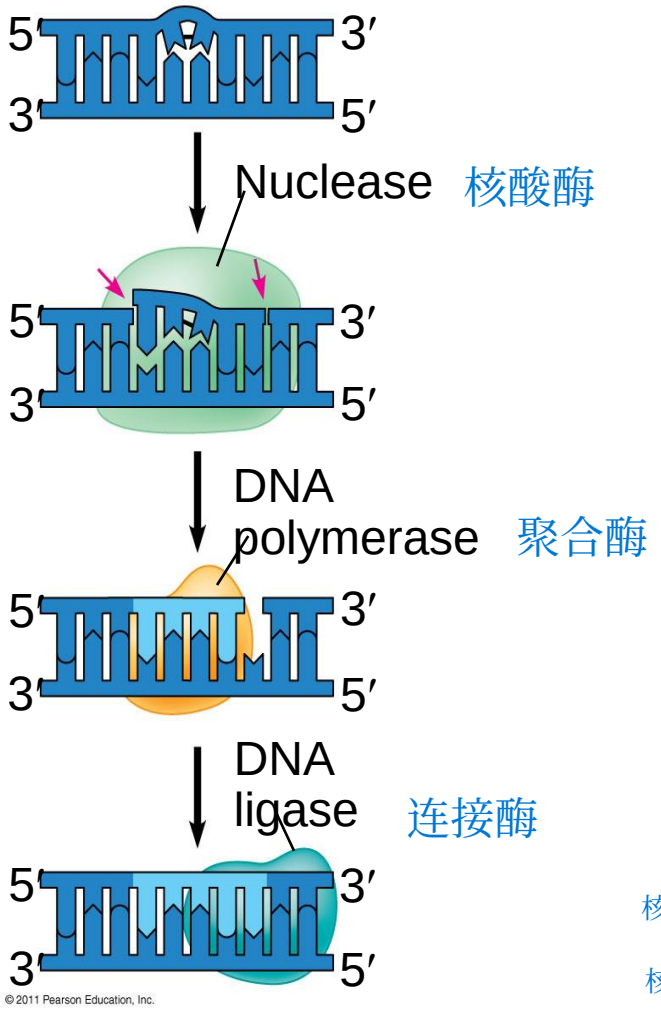
- The proteins that participate in DNA replication form a large complex, a "DNA replication machine"
- The DNA replication machine may be stationary during the replication process
- Recent studies support a model in which DNA polymerase molecules "reel in" 卷入 parental DNA and "extrude" newly made daughter DNA molecules

校对和修复DNA
Proofreading and Repairing DNA

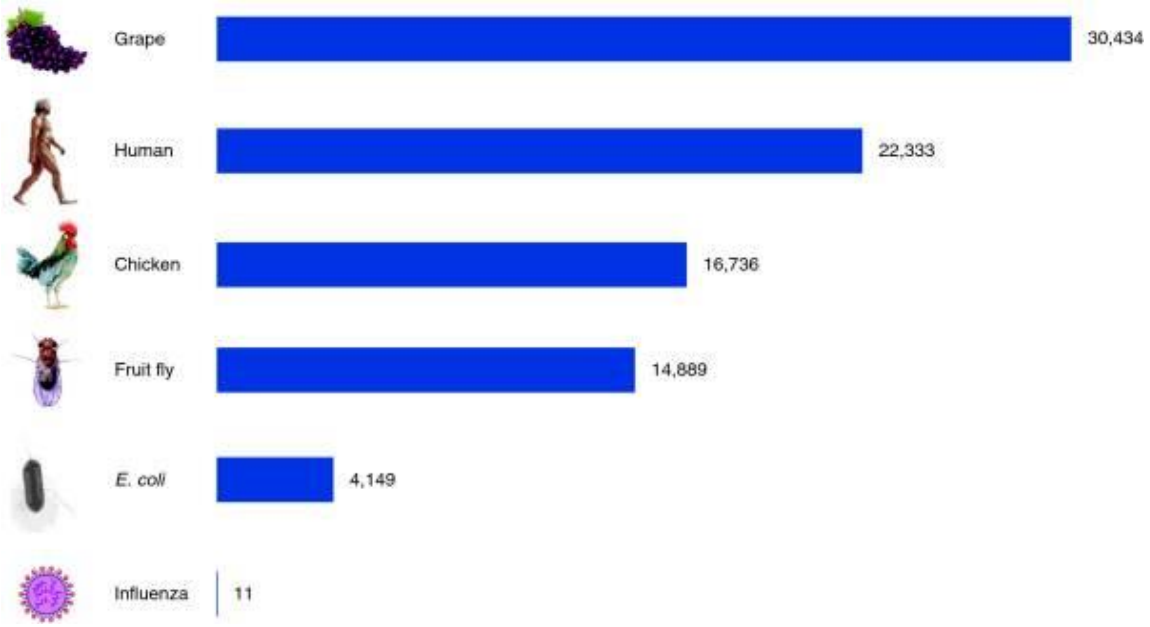


- Mismatch DNA structure can be recognized by nuclease. 核酸内切酶
- DNA polymerases proofread newly made DNA, replacing any incorrect nucleotides (from one in 10^7 to one in 10^9)
- In **mismatch repair** of DNA, repair enzymes correct errors in base pairing
- DNA can be damaged by exposure to harmful chemical or physical agents such as oxidants, ultraviolet light and X-rays; it can also undergo spontaneous changes. (1million damages/cell/day)
- In **nucleotide excision repair**, a **nuclease** cuts out and replaces damaged stretches of DNA

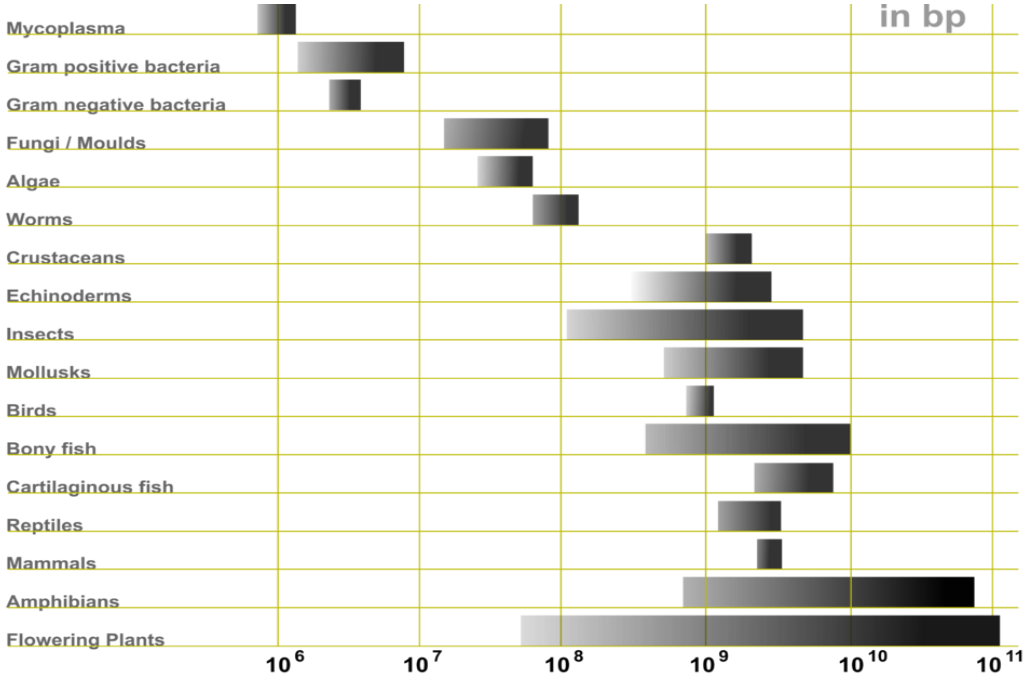
核酸内切酶
- Endonuclease: cut nucleotides in middle of a DNA

核酸外切酶
- Exonuclease: cut nucleotides the end of a DNA

Genome size 基因组大小



Gene counts in a variety of species



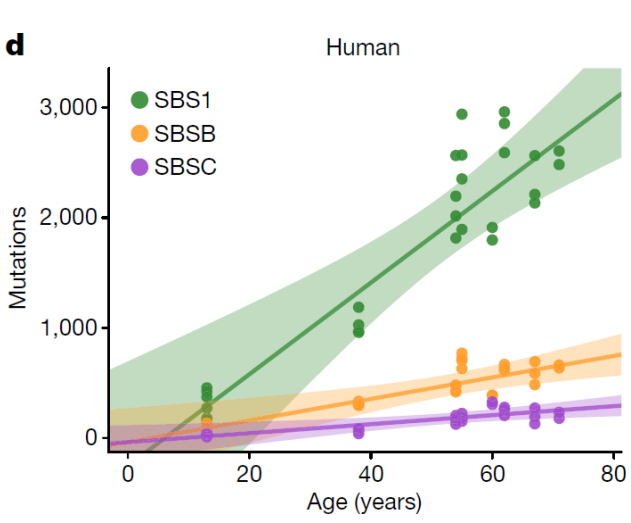
Genome size
Am.J. of Botany (2018)

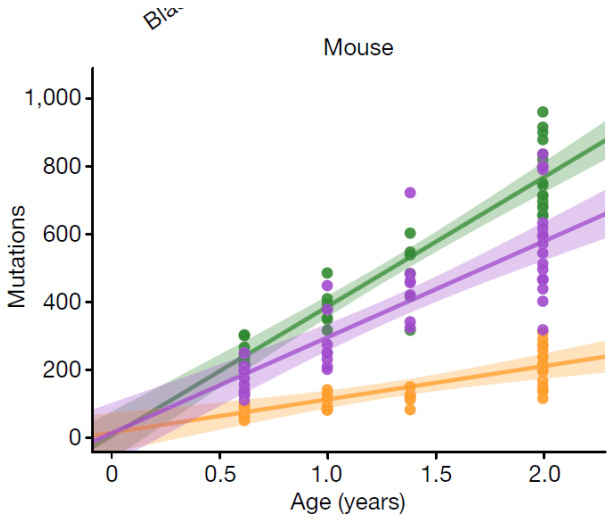
Having a large genome increases the risk of extinction (grow more slowly) 拥有庞大的基因组会增加灭绝的风险。（增长得更慢）

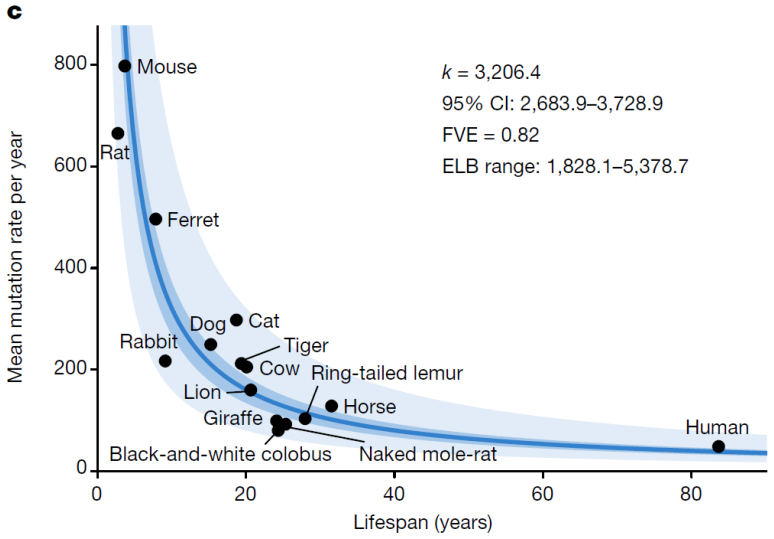
Evolutionary Significance of Altered DNA Nucleotides

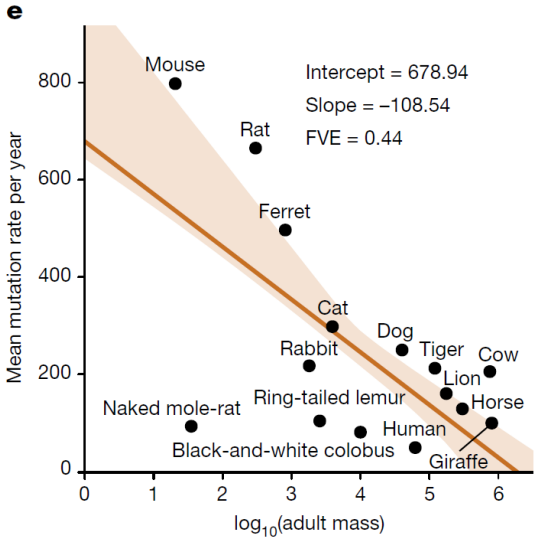
- Error rate after proofreading repair (10^9) (6x 10^9 bs/Cell) is low but not zero
- Sequence changes may become permanent and can be passed on to the next generation
- These changes (mutations) are the source of the genetic variation upon which natural selection operates
- Sperm may carry 50-500 mutations
- Champs and human 2% differences.
- Among us, we have 0.1~0.4% difference in the noncoding (intron regions) (our individuality come from this difference)
- Do we have Selfish gene (lazy gene, sorry gene, self-esteem gene….) ?
- DNA from bone has a half-life of 521 years

Mutation rate and aging 突变率和衰老









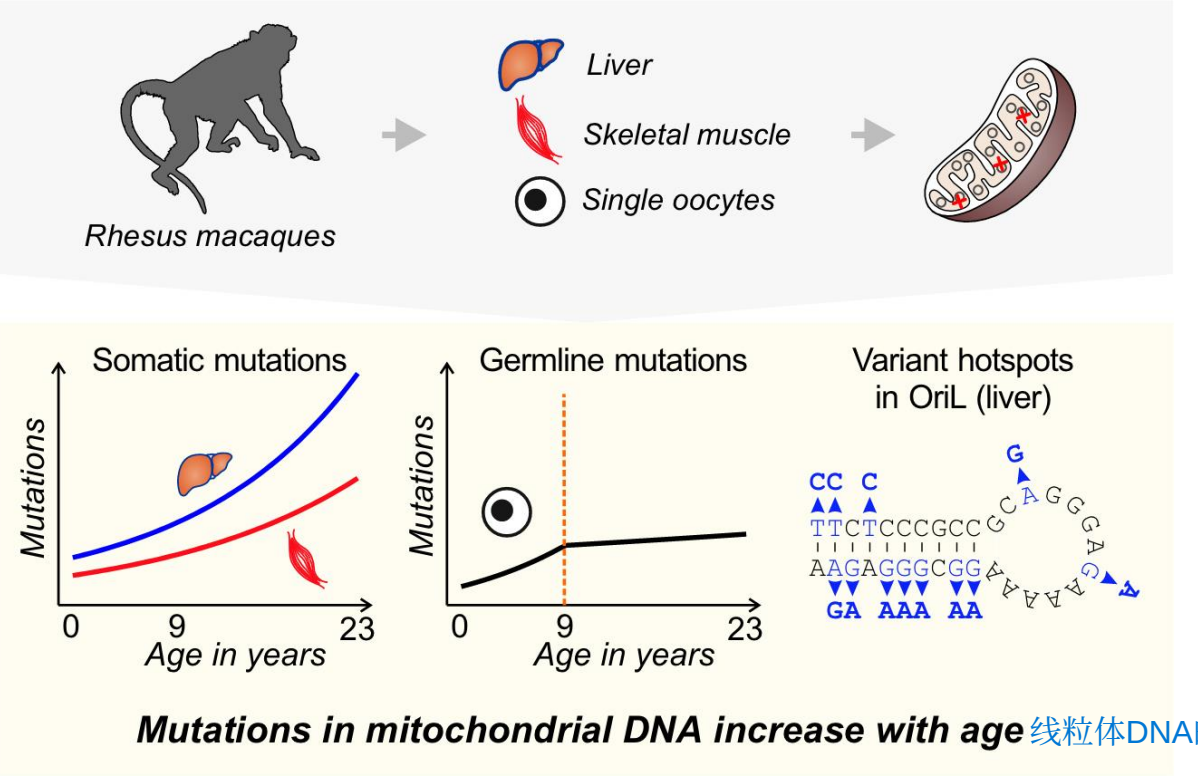
Mutagen	Risk factors	Health behavior that prevents exposure
Sunlight	Increased exposure to sunlight leading to thymine dimers in DNA	• Wear sunscreen • Avoid excess sun exposure • Reapply sunscreen every 2 h • Wear clothes/hat that cover skin
X-ray radiation	• Multiple X-rays • Close proximity to X-rays	• Limit the number of X-rays to only ones that are necessary • Wear protective lead vest over areas not being X-rayed • Use alternative imaging devices
Tobacco products	• Smoking • Chewing tobacco • Second-hand smoke	• Avoid smoking/tobacco products • Avoid being near smoke
Chemicals	Increased exposure to carcinogenic/mutagenic chemicals	• Limit exposure • Dispose of chemicals properly • Wear protective gear
Nitrites $[O{=}N{=}O]^-$	Increased consumption of processed meats, such as hot dogs, sausages, and bacon	Limit consumption of processed meats and grilled meats

Nature 2022

30years

4 years

Somatic mutations 体细胞突变



Mutations in mitochondrial DNA increase with age 线粒体DNA的突变随着年龄增长而增加

- Somatic mutations can occur in any of the cells of the body except the germ cells (sperm and egg) and therefore are not passed on to children.

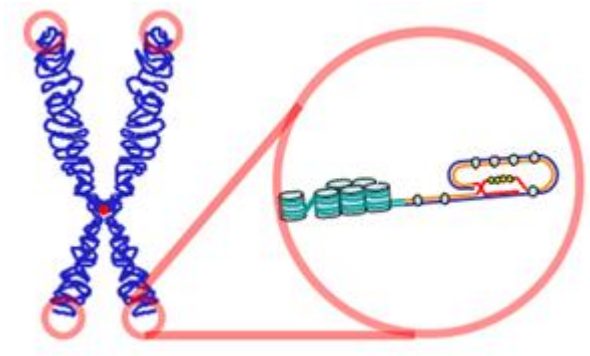
体细胞突变可发生在除生殖细胞**（精子和卵子）**以外的任何体细胞中，因此不会遗传给后代。

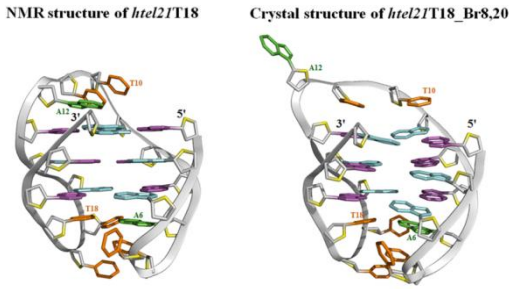
Dolly (5 July 1996 – 14 February 2003) was a female domestic sheep, and the first mammal cloned from an adult somatic cell, using the process of nuclear transfer. (normally, 10 – 12 years. Telomere shortening ?)

- Human DNA is 98 percent identical to chimpanzee DNA. Human DNA is 30% identical to *E. coli* DNA.
- The average amount of genetic difference between any two humans is 0.2 %, or one in 500 bases. Chimpanzees differ by 0.8 %.

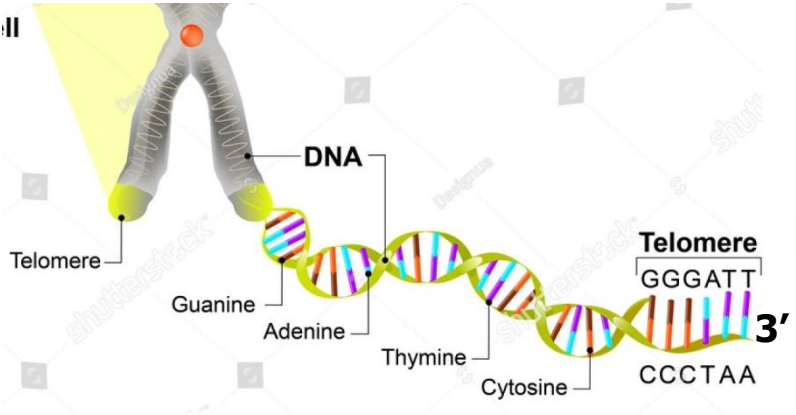
端粒在真核染色体中
Telomeres in Eukaryotic Chromosome

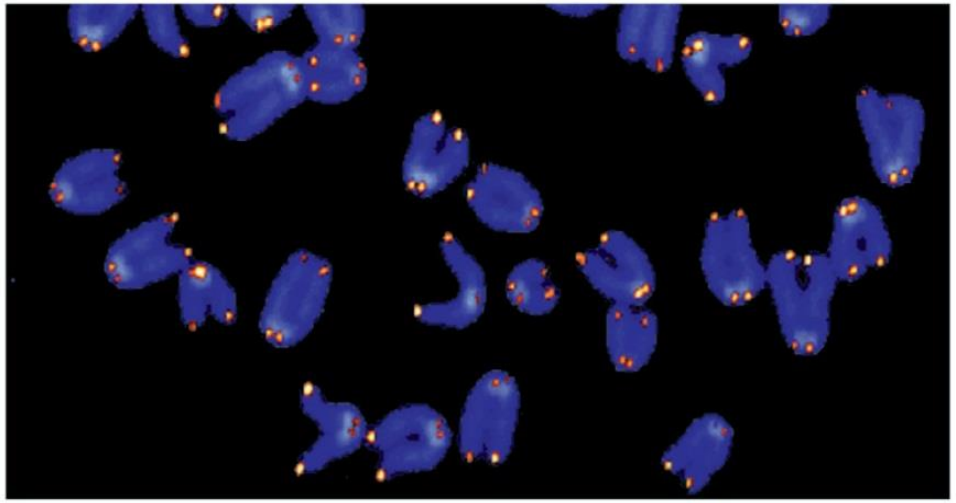
Telomere Replication.mp4





Liu et al (2019) NAR; Geng et al (2019) Chem Sci.

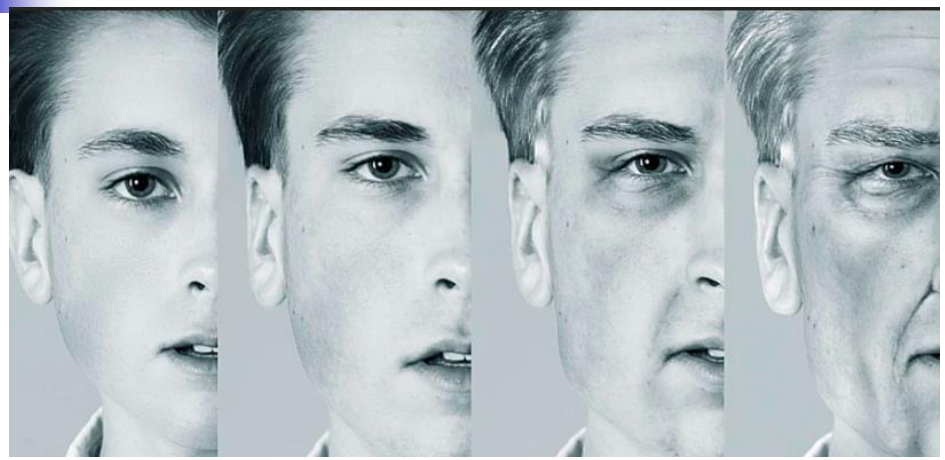


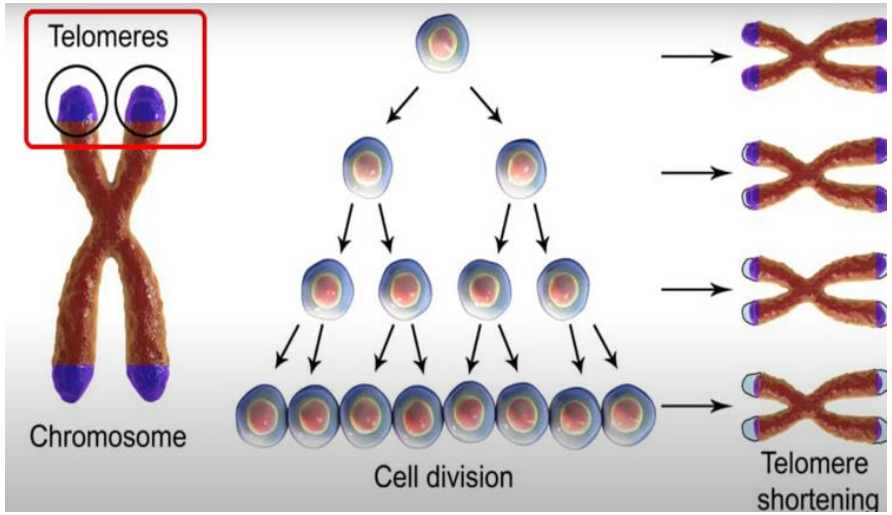


- Eukaryotic chromosomal DNA molecules have special nucleotide sequences at their ends called **telomeres**
- Telomeres do not prevent the shortening of DNA molecules, but they do postpone the erosion of genes near the ends of DNA molecules

真核生物染色体的 DNA 分子在其末端具有特殊的核苷酸序列，称为端粒
端粒并不能阻止 DNA 分子的缩短，但它们确实能延缓 DNA 分子末端附近基因的侵蚀

Human Aging





- If you freeze a cell at G1 in 20th cycle, and then it is thawed, the cell cycle will start at 20th cycle in G1 phase! (you can prolong the life of a cell)
- **Hayflick limit**: a normal human cell population will divide between 40 and 60 times in cell culture before entering a senescence phase (G0 phase) 衰老期
- It has been proposed that the shortening of telomeres is connected to aging (Vitamin D can increase it) 端粒酶
- An enzyme called **telomerase** catalyzes the lengthening of telomeres in germ cells (not in other normal cells)
- Telomerase contains an RNA primer for the extension of or the synthesis of new telomere in the 3’ end before DNA replication

高压氧治疗

Hyperbaric Oxygen Therapy

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A new study from Tel Aviv University (TAU) and the Shamir Medical Center in Israel indicates that hyperbaric oxygen treatments (HBOT) in healthy aging adults can stop the aging of blood cells and reverse the aging process. In the biological sense, the adults' blood cells actually grow younger as the treatments progress.

“Hyperbaric oxygen therapy increases telomere length and decreases immunosenescence in isolated blood cells: a prospective trial” Aging on November 18, 2020.19 Jan 2021

Biotechnology

Example of MC

(46) **Transport protein on cell membrane can**

(a) help charged molecules passing through cell.

(b) help hydrophobic molecules passing through cell.

(c) help both charged and hydrophobic molecules passing through cell.

(d) None of the above.

Answer (a)

Each paper will be graded by three TAs

Final exam: Open book, multiple choices

Topics for SCIE1120 term paper (May 29 at 10:00am. 2025, submission June. 4)

(1) Origin of life
(2) The Cambrian Explosion
(3) Detection of Covid-19 by RT-PCR
(4) Anti-covid-19 mRNA vaccine
(5) Mitochondria and related human diseases
(6) Insulin and related human diseases
(7) Telomere and human aging
(8) DNA mutations and related human diseases
(9) Stem cell and human health
(10) Collagen and human diseases
(11) Cellular senescence and human health
(12) Gene editing and human health
(13) Liposome and drug delivery
(14) How could supplements Nicotinamide Mono-nucleotide (NMN) and Nicotinamide Riboside (NR) extend human health and life span and what are their side-effects?

Report

We will have 14 topics for the term papers. The topics will be announced on Canvas. There are a maximum of 14 students per topic. After 14 students have chosen a topic, it will be closed, please select another topic.

Generally, the paper should have an introduction, background, application (or scientific importance) and conclusion/summary sections according to your ideas and topic. It can also have figures (which will not count towards the size), you can also number the references in the main text ([1], [2], [3]...). The term paper should be around 5-6 pages Times 12pt single spaced and contain at least 5 references (not included in the 5-6 pages). Any significantly shorter or longer essay will get a deduction in grade. Please note that you should formulate in your own words, any content / sentence / figure that is not from you needs to be referenced. I will open an assignment in Canvas for submission of the paper later.

Marking is based on:

Content and Scientific Accuracy and Significance (30%)
Content refers to a good description and analysis of the topic. Scientific Accuracy refers to proper description, implementation, and elaboration of scientific concepts as appropriate for the topic.
Mechanism and Depth (50%)
Development of concepts. Evidence of background reading and research. Presentation of different theories and viewpoints as appropriate for the topic. Use of appropriate references where relevant. (Including biomolecules involved)
Organization and Style (10%)
Appropriate and logical structure both within the term paper as a whole and within subsections and paragraphs.
Grammar (10%)